Monday Feb-24-2025

Flowserve Facility:

Flowserve Nordstrom / Valbart Valves
1511 Jefferson St.
Sulphur Springs, TX 75482

cmtrs@flowserve.com

To:

CERTIFICATE OF CONFORMANCE

SERIAL #	HEAT #	PO #	Description	Class	End Connection
61-618460	1347S	STP46777	4" 2345	900	FLG
61-618461	1347S	STP46777	4" 2345	900	FLG
61-618462	1348S	STP46777	4" 2345	900	FLG
61-618463	1348S	STP46777	4" 2345	900	FLG
61-619325	1358S	STP46777	4" 2345	900	FLG
61-619326	1347S	STP46777	4" 2345	900	FLG
61-619327	1358S	STP46777	4" 2345	900	FLG
61-619328	1369S	STP46777	4" 2345	900	FLG
61-619453	1369S	STP46777	4" 2345	900	FLG
61-619454	1330S	STP46777	4" 2345	900	FLG
61-619455	1330S	STP46777	4" 2345	900	FLG
61-619456	2038S	STP46777	4" 2345	900	FLG
61-619498	2023S	STP46777	4" 2345	900	FLG
61-619499	2019S	STP46777	4" 2345	900	FLG
61-619500	2023S	STP46777	4" 2345	900	FLG

We hereby certify that all the subject valves/parts/sealant have been manufactured in conformance with the applicable requirements of Flowserve Sulphur Springs Operation (FSSO) specifications and that the valve/parts/sealant materials for component parts meet the applicable requirements of FSSO drawings and ASTM specifications, ASME B16.34 (If Applicable), API Q1 10th edition, and API 6D 25th edition.(if applicable to the API monogram). Flowserve Dynamic Balanced Valves are provided with a Pressure balance feature unless specifically indentified in the Valve description. All valves/parts/sealant meet all the requirements of the customer purchase order.

Documentation Technician
Justin McKee

FORM QF-MT VER 4

FLOW LINK SYSTEMS (P) LTD FOUNDRY UNIT - II 65 / 2-A,B & C,PUDUPALAYAM B.P.O AVINASHI Tk, COIMBATORE - 641 654 INDIA	**TEST CERTIFICATE** (EN 10204-3.1)	TC NO : R-3838 DATE : 16-Oct-18
CUSTOMER FLOWSERVE INDIA CONTROLS PVT LTD, B8, MMDA Industrial Area, Maraimalai Nagar-603209 Kancheepuram, INDIA	SPECIFICATION REFERENCE 1. ASTM A216-18 Gr.WCC 2. Customer Spec. RMC 90012 Rev. 13	HEAT NO. 1330S Date of Pouring 7-Jul-18 FOUNDRY MARK FLS

PO NUMBER & DATE	ORDER ACCEPTANCE NO
10082358 DT: 15.05.2018	18-0773

CHEMICAL COMPOSITION % wt

MELTING PRACTICE : Steel melted using Electric Induction furnace, and fully killed using Aluminium.

SPECIFICATION	C	Si	S	P	Mn	Ni	Cr	Mo	V	Cu	Al	--	*RE	**CE
Minimum	--	--	--	--	--	--	--	--	--	--	--	--	--	--
Maximum	0.23	0.60	0.020	0.025	1.20	0.50	0.50	0.20	0.030	0.30	0.08	--	1.00	0.43
Achieved	0.198	0.443	0.008	0.015	0.971	0.072	0.053	0.007	0.003	0.013	0.047	--	0.15	0.38

Heat Treatment : NORMALIZING: Temperature raised to 920°C, Soaked for 4 hours and then atmospheric air-cooled

Visual Inspection - Castings conforms to MSS-SP-55-2011 - Type II to XII (within level 'b')

MECHANICAL PROPERTIES

TEST TYPE	Tensile as per ASTM A370-17 / ~~ISO 6892-1-2016~~						Impact (CVN) ISO 148-2016/ASTM A370-17		BEND TEST		HARDNESS
SPECIFICATION	YS Rp 0.2% (KSI)	YS Rp 1.0% (KSI)	UTS (KSI)	% Elongation 4d	% Elongation 5d	% RA	Test Temperature (at °C) J Values	Avg.	ANGLE	D	(HBW / HRBW)
Minimum	40	--	70	22	--	35	--	--	--	--	--
Maximum	--	--	95	--	--	--	--	--	--	--	--
Achieved	49.4	--	79.5	33	--	66	--	--	--	--	155

DETAIL OF CASTINGS

S.No.	PO S.No.	DESCRIPTION	CAST DRAWING NO / REV	PART NO	POURED QTY	DESP. QTY
1	01	4" 900 FE BODY	D-487876 / S	N2757928-60	5	1

MTC - REVIEWED & ACCEPTED
10139322 FICPL QC

REMARKS:
1. Stated Chemical composition is from heat analysis; and Mechanical properties of test coupon from Cast keel block
2. *RE = Ni + Cr + Mo +V +Cu
3. $** \; CE = C + \frac{Mn}{6} + \frac{Cr+Mo+V}{5} + \frac{Ni+Cu}{15}$
4. The above statedKSI values have been calculated based on original tested N/MM² values.

CERTIFIED THAT
1. THE ABOVE GIVEN DETAILS ARE CORRECT, FULFILLS THE SPECIFICATION / STANDARDS AND PO.
2. MATERIAL WAS MANUFACTURED, SAMPLED, TESTED, INSPECTED IN ACCORDANCE WITH THE ABOVE SPECIFICATIONS AND CERTIFIED AS PER EN 10204 - 3.1.
3. THE MATERIAL IS FREE OF CONTAMINATION FROM RADIO ACTIVE ELEMENT OR RADIATION.

PREPARED BY:	CHECKED BY:	
S.MANIBHARATHI	R.SATHISH KUMAR ASSISTANT ENGINEER	M.S.DHANABALAN ASSISTANT MANAGER -QUALITY MANUFACTURER'S AUTHORIZED INSPECTOR